THE TATE CONJECTURE AS A PARAMETRIC PROPOSITION: A MACHINE-CHECKED LEAN 4 INTERFACE WITHOUT AXIOMS

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Abstract. We state the Tate conjecture in Lean 4 as a proposition that depends on an explicit structure `TateSetting`, in place of four user-declared axioms. The structure carries a type of hosts (smooth projective varieties over a finite field), types of codimension-i cycle classes and of Galois invariants in $H^{2i}_{\text{et}}(\overline{X}, \mathbb{Q}_l(i))$, and a cycle class map. For every such setting we prove that the statement is exactly surjectivity of the cycle class map, and that the fourfold-surfaces case is the codimension-2 statement. Neither the Tate conjecture nor any case of it is proved. The development builds with no `sorry`, no `native_decide` and no user axioms.

1. Introduction

Let X be a smooth projective variety over a finite field k, with $\overline{X}$ its base change to an algebraic closure and $l \neq \operatorname{char} k$. The Tate conjecture [1] asserts that the cycle class map

$$\mathrm{cl}: \ Z^i(X) \otimes \mathbb{Q}_l \to H^{2i}_{\text{et}}(\overline{X}, \mathbb{Q}_l(i))^{\mathrm{Gal}(\bar{k}/k)}$$

is surjective. It is open in general. The repository previously introduced the host type, the two coefficient types and the cycle class map as Lean axioms. In this version they are fields of a structure, so the conjecture is a proposition about an arbitrary setting supplied by the user, and the Lean environment contains no user axioms.

2. Definitions

Definition 2.1 (`TateConjecture.TateSetting`)**.** A *Tate setting* S consists of a type $\mathcal{X}$, types $C(X, i)$ and $G(X, i)$ for $X \in \mathcal{X}$ and $i \in \mathbb{N}$, and maps $\mathrm{cl}_{X,i} : C(X, i) \to G(X, i)$.

Definition 2.2 (`Statement`, `openSentence`, `fourfoldSurfaces`)**.** Statement(S, X, i) means: for every $y \in G(X, i)$ there is $x \in C(X, i)$ with $\mathrm{cl}_{X,i}(x) = y$. The proposition `openSentence` is defined to be Statement(S, X, i), and `fourfoldSurfaces` is Statement$(S, X, 2)$.

3. Results

Proposition 3.1 (`TateConjecture.statement_iff_surjective`)**.** *For every Tate setting S, host X and $i \in \mathbb{N}$,* Statement(S, X, i) *holds if and only if* $\mathrm{cl}_{X,i}$ *is surjective.*

Proposition 3.2 (`TateConjecture.fourfoldSurfaces_iff`)**.** *For every Tate setting S and host X,* `fourfoldSurfaces` *holds if and only if* Statement$(S, X, 2)$ *holds.*

Both hold by definition, and Lean proves them by `Iff.rfl`.

4. Correspondence between claims and Lean declarations

Every mathematical claim of this paper is listed below with the Lean declaration that proves it. "Conditional" means the hypothesis named in the statement is an explicit argument of the Lean theorem and is *not* proved in the repository. No claim in this paper goes beyond these statements.

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Claim	Lean declaration	Status
Statement ⇔ cycle class map surjective	`TateConjecture.statement_iff_surjective`	Proved (for every setting)
Fourfold-surfaces case = codimension 2	`TateConjecture.fourfoldSurfaces_iff`	Proved (for every setting)

5. Scope and limitations

Varieties over finite fields, étale cohomology, Galois actions and algebraic cycles are not formalized. A `TateSetting` may be any data at all, and the propositions above are unfolding identities. The Tate conjecture is not proved, in any codimension or for any variety. The relation of a Frobenius eigenvalue q^2 to cycles is not formalized either. Statement changes in this version: the axioms `SmoothProjOverFiniteField`, `CycleClasses`, `GalInvariants` and `clEtale` are now fields of `TateSetting`. `Statement`, `openSentence` and `fourfoldSurfaces` now take the setting as an extra argument. The two propositions above were added. The repository had no theorems before.

Appendix A. Lean source

The complete Lean source of the certified development (copyright headers omitted) is reproduced verbatim.

`TateConjecture/Basic.lean`.

```
/-!
# Tate conjecture: named sentence

Not a proof. Not Hodge. Not imported by BenFrohman/HODGE.
A Frobenius eigenvalue q^2 is a numerical condition, not a cycle.

The former axioms (`SmoothProjOverFiniteField`, `CycleClasses`, `GalInvariants`,
`clEtale`) are now fields of the structure `TateSetting`. Every definition and
theorem is parametric in such a setting; nothing about actual varieties is proved.
-/

namespace TateConjecture

/-- Abstract data for the Tate conjecture: placeholder hosts over a finite field,
cycle classes, Galois invariants in étale cohomology `H^{2i}(Q_ℓ(i))`, and a cycle
class map. Not schemes; supplied by the user of the statement. -/
structure TateSetting where
  SmoothProjOverFiniteField : Type
  CycleClasses : SmoothProjOverFiniteField → Nat → Type
  GalInvariants : SmoothProjOverFiniteField → Nat → Type
  clEtale : (X : SmoothProjOverFiniteField) → (i : Nat) → CycleClasses X i → GalInvariants X i

variable (S : TateSetting)

/-- Tate in codimension i: clEtale is surjective. Named, not proved. -/
def Statement (X : S.SmoothProjOverFiniteField) (i : Nat) : Prop :=
  ∀ y : S.GalInvariants X i, ∃ x : S.CycleClasses X i, S.clEtale X i x = y

/-- The open sentence. There is no term of this type in this repository. -/
def openSentence (X : S.SmoothProjOverFiniteField) (i : Nat) : Prop :=
```

```
  Statement S X i

/-- Fourfold surfaces: i = 2. Still a Prop. -/
def fourfoldSurfaces (X : S.SmoothProjOverFiniteField) : Prop :=
  Statement S X 2

/-- The statement is exactly surjectivity of the cycle class map. -/
theorem statement_iff_surjective (X : S.SmoothProjOverFiniteField) (i : Nat) :
    Statement S X i ↔ Function.Surjective (S.clEtale X i) := Iff.rfl

/-- The fourfold-surfaces sentence is the codimension-2 statement. -/
theorem fourfoldSurfaces_iff (X : S.SmoothProjOverFiniteField) :
    fourfoldSurfaces S X ↔ Statement S X 2 := Iff.rfl

end TateConjecture
```

Appendix B. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/TateConjecture`
- Verified commit:
  `a308b72ab10e1b7f7670df1b5c1288cdb2e783db`
- Lean toolchain: `leanprover/lean4:v4.34.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/TateConjecture/actions/runs/41794909195`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/TateConjecture/archive/a308b72ab10e1b7f7670df1b5c1288cdb2e783db.tar.gz`
  `cf8e038db9227fe97bc5a8fb0c0fb43fdb1429d20a43bdc3438cb20b6e79d8d8`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 2 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`TateConjecture.fourfoldSurfaces_iff`	`(no axioms)`
`TateConjecture.statement_iff_surjective`	`(no axioms)`

References

[1] J. Tate, *Algebraic cycles and poles of zeta functions*, in: Arithmetical Algebraic Geometry, Harper & Row (1965), 93–110.

[2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

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URL: `https://github.com/BenFrohman/TateConjecture`